

1D Heat Conduction (FTCS): Neumann and Convection Boundary Conditions

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Where we are

- Governing PDE (constant properties, 1D): $\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$
- Spatial grid $x_i = i\Delta x$, time levels $t^n = n\Delta t$
- FTCS interior update:

$$T_i^{n+1} = T_i^n + \lambda (T_{i+1}^n - 2T_i^n + T_{i-1}^n), \quad \lambda = \frac{\Delta t}{\Delta x^2}$$

- Today: **Neumann** (specified gradient/flux) and **Convection** (Robin) BCs using **ghost nodes**.

1D fin and boundary conditions

Left boundary ($x = 0$)

Dirichlet: $T_0 = \bar{T}$

Neumann: $T'_0 = q_0$

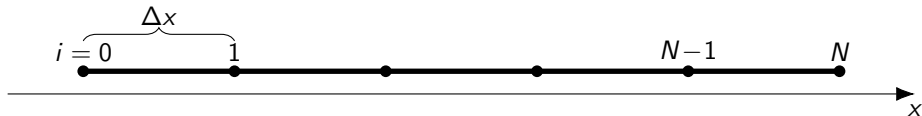
Convection: $-kT'_0 = h(T_0 - T_\infty)$

Right boundary ($x = L$)

Dirichlet: $T_N = \bar{T}$

Neumann: $T'_L = q_L$

Convection: $-kT'_L = h(T_N - T_\infty)$



- Interior nodes: $i = 1, \dots, N - 1$
- Goal: derive explicit time updates for T_0 and T_N

Boundary update at x_0 using a 3-term Taylor series (no ghost node)

Start with a three-term Taylor series about x_0 :

$$T(x) = T_0 + (x - x_0) T'_0 + \frac{(x - x_0)^2}{2} T''_0 + O((x - x_0)^3).$$

Evaluate at $x_1 = x_0 + \Delta x$:

$$T_1 = T_0 + \Delta x T'_0 + \frac{\Delta x^2}{2} T''_0 + O(\Delta x^3).$$

Solve for T''_0 :

$$\boxed{T''_0 \approx \frac{2}{\Delta x^2} (T_1 - T_0 - \Delta x T'_0)} \quad (\text{second-order in space}).$$

Now substitute into the PDE at x_0 :

$$\left. \frac{\partial T}{\partial t} \right|_0 = \alpha T''_0 \quad \Rightarrow \quad \frac{T_0^{n+1} - T_0^n}{\Delta t} = \frac{2\alpha}{\Delta x^2} (T_1^n - T_0^n - \Delta x T_0'^n).$$

Neumann BC at the left end: $T'_0 = q_0$

So the explicit FTCS-style boundary update is:

$$T_0^{n+1} = T_0^n + 2\lambda (T_1^n - T_0^n - \Delta x T_0'^n), \quad \lambda = \frac{\alpha \Delta t}{\Delta x^2}.$$

Neumann BC: if $T'_0 = q_0$, then simply set $T_0'^n = q_0$ in the update. Neumann BC (specified gradient):

$$\left. \frac{\partial T}{\partial x} \right|_{x=0} = -\frac{q_0}{k}.$$

Neumann BC at $x = L$: specified heat flux q_L (no ghost node)

At the right boundary $x_N = L$, use a three-term Taylor series about x_N :

$$T(x) = T_N + (x - x_N) T'_N + \frac{(x - x_N)^2}{2} T''_N + O((x - x_N)^3).$$

Evaluate at the interior node $x_{N-1} = x_N - \Delta x$:

$$T_{N-1} = T_N - \Delta x T'_N + \frac{\Delta x^2}{2} T''_N + O(\Delta x^3).$$

Solve for T''_N :

$$T''_N \approx \frac{2}{\Delta x^2} (T_{N-1} - T_N + \Delta x T'_N) \quad (\text{second-order in space}).$$

Neumann flux condition at $x = L$:

$$q_L = -k T'_N \quad \Rightarrow \quad T'_N = -\frac{q_L}{k}.$$

Substitute into the PDE at x_N using FTCS in time:

Robin (Convection) BC at $x = 0$

Boundary condition (Newton's law of cooling):

$$-k T'_0 = h (T_0 - T_\infty)$$

Start with 3-term Taylor series about x_0 :

$$T_1 = T_0 + \Delta x T'_0 + \frac{\Delta x^2}{2} T''_0$$

Solve for curvature:

$$T''_0 \approx \frac{2}{\Delta x^2} (T_1 - T_0 - \Delta x T'_0)$$

Robin BC at $x = 0$: Explicit Time Update

From the boundary condition:

$$T'_0 = -\frac{h}{k}(T_0 - T_\infty)$$

Insert into curvature:

$$T''_0 = \frac{2}{\Delta x^2} \left[T_1 - T_0 + \Delta x \frac{h}{k} (T_0 - T_\infty) \right]$$

Apply FTCS in time:

$$T_0^{n+1} = T_0^n + 2\lambda [(T_1^n - T_0^n) + \text{Bi}(T_0^n - T_\infty)]$$

where

$$\text{Bi} = \frac{h\Delta x}{k}, \quad \lambda = \frac{\alpha\Delta t}{\Delta x^2}$$

Robin (Convection) BC at $x = L$

Boundary condition:

$$-k T'_N = h(T_N - T_\infty)$$

Taylor expansion about x_N :

$$T_{N-1} = T_N - \Delta x T'_N + \frac{\Delta x^2}{2} T''_N$$

Solve for curvature:

$$T''_N \approx \frac{2}{\Delta x^2} (T_{N-1} - T_N + \Delta x T'_N)$$

Robin BC at $x = L$: Explicit Time Update

From the boundary condition:

$$T'_N = -\frac{h}{k}(T_N - T_\infty)$$

Insert into curvature:

$$T''_N = \frac{2}{\Delta x^2} \left[T_{N-1} - T_N - \Delta x \frac{h}{k} (T_N - T_\infty) \right]$$

FTCS time update:

$$T_N^{n+1} = T_N^n + 2\lambda \left[(T_{N-1}^n - T_N^n) - \text{Bi}(T_N^n - T_\infty) \right]$$

$$\text{Bi} = \frac{h\Delta x}{k}$$

FTCS Algorithm with General Boundary Conditions

Given:

$$N, \Delta x, \Delta t, \alpha, T_i^0 \quad \lambda = \frac{\alpha \Delta t}{\Delta x^2}$$

For each time step $n \rightarrow n + 1$:

1. Interior nodes ($i = 1, \dots, N - 1$)

$$T_i^{n+1} = T_i^n + \lambda (T_{i+1}^n - 2T_i^n + T_{i-1}^n)$$

2. Left boundary update

$$\text{Dirichlet: } T_0^{n+1} = \bar{T}$$

$$\text{Neumann: } T_0^{n+1} = T_0^n + 2\lambda (T_1^n - T_0^n - \Delta x q_0)$$

$$\text{Robin: } T_0^{n+1} = T_0^n + 2\lambda [(T_1^n - T_0^n) + \text{Bi}(T_0^n - T_\infty)]$$

Right Boundary Updates

Right boundary ($x = L$)

Dirichlet: $T_N^{n+1} = \bar{T}$

Neumann (flux q_L) :

$$T_N^{n+1} = T_N^n + 2\lambda \left(T_{N-1}^n - T_N^n - \Delta x \frac{q_L}{k} \right)$$

Robin:

$$T_N^{n+1} = T_N^n + 2\lambda \left[(T_{N-1}^n - T_N^n) - \text{Bi}(T_N^n - T_\infty) \right]$$

$$\text{Bi} = \frac{h\Delta x}{k}$$

Stability (Explicit Diffusion)

Interior FTCS stability condition:

$$\lambda = \frac{\alpha \Delta t}{\Delta x^2} \leq \frac{1}{2}$$

- Boundary updates do not change the basic diffusion stability limit.
- Smaller $\Delta x \Rightarrow$ smaller allowable Δt .
- Copper (α large) $\Rightarrow$ stricter time-step restriction.

Computational insight: Explicit diffusion is conditionally stable — time step controlled by grid spacing.

Summary

- Use a 3-term Taylor series at the boundary node.
- Solve for T'' directly from interior information.
- Substitute boundary condition into the curvature expression.
- Result: explicit time updates for T_0 and T_N .

Boundary types:

Dirichlet: prescribe T

Neumann: prescribe T'

Robin: prescribe $-kT' = h(T - T_\infty)$

All three fit naturally into the FTCS update structure.